KiwiSaver Government Contribution

Microsimulation Model: Methodology and Data Sources

1. Purpose and Scope

This document describes the methodology, data sources, and key assumptions underpinning the KiwiSaver Government Contribution (GVC) Microsimulation Model. The model is developed by Te Ara Ahunga Ora Retirement Commission to:

- Reproduce the observed 2024 GVC fiscal cost as a validated baseline
- Simulate the fiscal cost of the 2026 GVC policy settings across the KiwiSaver member population
- Allow policy analysts to explore the sensitivity of fiscal cost to changes in match rate, maximum contribution, minimum contribution rate, income targeting, age targeting levers income cap, age eligibility, contribution rate uptake, and income growth.

The model should be interpreted alongside the documented assumptions in Section 6.

2. Policy Settings Modelled

2.1 2024 Policy (Observed Baseline)

The 2024 policy settings reflect the GVC prior to the Budget 2025 changes:

Parameter	2024 Value
Maximum annual government contribution	$521.43
Match rate	$0.50 per $1 of member contribution
Contribution needed to receive full GVC	$1,042.86 per annum
Income at which full GVC is reached (3% contribution rate)	~$34,762
Eligible age range	18–65 (pro-rata at boundaries)
Income eligibility cap	None
Minimum employee contribution rate	3% (voluntary for self-employed)

The 2024 baseline is drawn directly from observed IRD administrative data and involves no estimation or simulation. The total GVC paid in 2024 was $1,022,341,210 across 3,348,048 KiwiSaver members.

2.2 2026 Policy (Simulation)

The 2026 settings represent the Government Contribution (GVC) after the Budget 2025 changes, but before the contribution rate increase to 4% from 2028. Default simulation parameters are:

Parameter	2026 Default	User-adjustable
Maximum annual government contribution	$260.72	Yes (numeric input)
Match rate	$0.25 per $1 of member contribution	Yes (slider: 0.25–2.00, step 0.25)
Contribution needed to receive full GVC	$1,042.88 per annum	Derived (max GVC ÷ match rate)
Income at which full GVC is reached (3.5% rate)	~$29,797	Derived
Eligible age range	16–17 through 55–64 (default)	Yes (per-band checkboxes)
Income eligibility cap	$180,000	Yes ($10k - $180k stepped slider + “no cap” toggle)
Minimum employee contribution rate	3.5%	Yes (slider: 2%–10%)

The GVC formula applied in the simulation is:

```
GVC = min(member_income × member_rate × match_rate, max_gvc)
```

Members earning above the income eligibility cap, or outside the eligible age range, receive $0 GVC (eligibility is assessed on income, including any growth).

Age eligibility is set by ticking individual age bands (Under 16, 16–17, 18–24, 25–34, 35–44, 45–54, 55–64, 65+) rather than dragging a single min–max age slider. Each band is included or excluded in full. The underlying data does not support partial-band inclusion, so there is no equivalent of setting an exact age threshold like 17.5. Bands need not be contiguous, which supports age-cohort targeting scenarios (e.g. boosting 18–24 and 45–54 while leaving 25–44 untouched. See Section 5.1).

3. Data Sources

3.1 Source 1: Inland Revenue KiwiSaver GVC Administrative Data (2024)

Field	Detail
File	2024_IRD_Administrative_KiwiSaver_Data.xlsx
Provider	Inland Revenue (IRD)
Reference period	2024 tax year
Coverage	All 3,348,048 KiwiSaver members as at 30 June 2024
Used for	2024 observed baseline only

This file contains pre-aggregated cross-tabulations showing member counts and total GVC paid by income band (8 bands), age band (7 bands), GVC amount band (9 bands), and gender. The average GVC per income band and per age band are derived directly from these totals.

Note: The 2024 IRD data uses coarser income bands ($20k wide, capped at $120k+) than the 2025 monitoring data. The 2024 baseline uses these observed averages directly rather than applying a formula, ensuring the model exactly reproduces the $1,022,341,210 total.

3.2 Source 2: Inland Revenue KiwiSaver Monitoring Project Data (2025)

Field	Detail
File	IRD_KiwiSaver_Monitoring_Project_Data_at_30_June_2025.xlsx
Provider	Inland Revenue (IRD)
Reference period	As at 30 June 2025; income data from 2024 tax year
Coverage	All 3,405,406* KiwiSaver members as at 30 June 2025
Used for	2026 simulation: age band member totals and contribution rate mix

* 173 of whom are excluded from the simulation grid; see Section 7.3

Key tables used from this file:

- Sheet 2: Member counts by age band × contribution rate. Primary source for KiwiSaver member totals per age band
- Sheet 6: Member counts by income band × PAYE contribution rate ($10k bands). Used for rate proportions by income band
- Sheet 14: Contribution status by income type. Primary source for true contributor/non-contributor split
- Sheet 15: Member counts by income type × contribution rate. Used to identify members with a captured rate vs unmatched contributors.

Critical distinction: Sheet 15 records members with an identifiable contribution rate captured as at 30 June 2025, a point-in-time snapshot. Sheet 14 records actual contribution status for the full year, an outcome measure. These two sources differ by 530,592 members, correctly treated as contributors in the model even though the snapshot didn't catch them; see Section 4.5 for the income-type breakdown of this group (a mix of PAYE members the snapshot missed and genuinely non-PAYE contributors, not exclusively the latter).

3.3 Source 3: IRD Taxable Income Distribution by Age (2025)

Field	Detail
File	IRD_Taxable_income_distribution_of_individuals_2025.xlsx
Provider	IRD
Reference period	2025
Coverage	Income-earning population of New Zealand: 4,689,830 people
Used for	Shape of income distribution within each age band (fine $1,000 bands)

This file contains the income distribution of the New Zealand income-earning population at $1,000-wide income bands, cross-tabulated by 14 age groups. It provides the within-band income distribution used to allocate KiwiSaver members across fine income bands. Member totals are anchored to the monitoring data (Source 2), not derived from this file.

> *This data is used only as a shaping tool — it determines how members are distributed across fine income bands within each age group. The total members per age band come directly from the KiwiSaver monitoring data (Sheet 2), not from this file.*

4. Model Architecture

4.1 2024 Baseline: Two Lookup Tables

The 2024 baseline is stored in two separate lookup tables, each independently reproducing the observed $1,022,341,210 total GVC:

- dataset_by_income.csv: 8 rows, one per IRD income band. Average GVC per member taken directly from the IRD spreadsheet.
- dataset_by_age.csv: 7 rows, one per age band. Average GVC per member derived as IRD observed total GVC ÷ observed member count per age band.

4.2 2026 Simulation: Fine-Grained Grid

The 2026 simulation uses a grid of 1,456 rows (182 income bands × 8 KiwiSaver age bands). Each row contains:

- members: KiwiSaver member count, scaled to match monitoring data exactly
- inc_low, inc_high, inc_mid: income band boundaries and midpoint
- pp3, pp4, pp6, pp8, pp10, ppn: revised contribution rate proportions

4.2.1 Member Count by Age Band

Member counts in the simulation grid are derived in two steps:

Step 1: Anchor to observed KiwiSaver totals. For each KiwiSaver age band, the total member count is taken directly from Sheet 2 of the 2025 monitoring data. These are the exact observed figures with no approximation.

KiwiSaver age band	Taxable Income equivalent	KS members (Sheet 2)
Under 16	u15 + 1/5 of 15–19	113,884
16–17	2/5 of 15–19	55,603
18–24	2/5 of 15–19 + all 20–24	393,184
25–34	25–29 + 30–34	740,046
35–44	35–39 + 40–44	736,764
45–54	45–49 + 50–54	590,744
55–64	55–59 + 60–64	537,553
65+	65–69 + 70–74 + 75+	237,455
Total	—	3,405,233

Step 2: Apply taxable income shape. For each age band, members are distributed across the 182 fine income bands using IRD population counts as a proportional shape, scaled so the total exactly matches the monitoring data:

```
scale_factor(age) = KS_actual(age) / inc_dist_total(age)
```

```
members(age, income) = inc_dist(age, income) × scale_factor(age)
```

This ensures the income distribution within each age band follows the taxable income shape, while the total per age band exactly matches the observed KiwiSaver membership. The taxable income distribution's $1,000-wide income bands also provide a clean boundary at the $180,000 income cap.

> *The IRD taxable income data includes an age band span of 15–19. Equal weighting per year (1/5) is assumed to split it into the KiwiSaver under-16, 16–17, and 18–24 bands. This is Assumption A4 in Section 6.*

4.2.2 Member Count by Broad Income Bands

For reference, the table below shows the same 3,405,233 KiwiSaver members grouped into the reporting income bands used in the tool's "By Income Band" view. These run in uniform $10,000 steps to $180,000, plus a single "Over $180k" group, a coarser grouping than the 182 fine bands used internally.

Broad income band	KiwiSaver members	% of total
$1 - $10k	527,980	15.5%
$10k - $20k	249,794	7.3%
$20k - $30k	443,444	13.0%
$30k - $40k	273,306	8.0%
$40k - $50k	238,991	7.0%
$50k - $60k	266,922	7.8%
$60k - $70k	272,678	8.0%
$70k - $80k	232,844	6.8%
$80k - $90k	178,668	5.2%
$90k - $100k	140,182	4.1%
$100k - $110k	113,413	3.3%
$110k - $120k	85,793	2.5%
$120k - $130k	67,689	2.0%
$130k - $140k	52,937	1.6%
$140k - $150k	42,800	1.3%
$150k - $160k	35,010	1.0%
$160k - $170k	28,620	0.8%
$170k - $180k	28,664	0.8%
Over $180k	125,497	3.7%
Total	**3,405,233**	**100.0%**

> *This is the whole population (all 8 age bands), consistent with the age-band table's convention of showing total KiwiSaver membership rather than GVC-eligible membership. The "Over $180,000" figure of 125,497 is the whole population total; the age-eligible-only figure used in Section 7.4's context material is smaller (120,426), reflecting the age exclusion discussed in Section 7.3, not an inconsistency between the two.*

4.2.3 GVC Calculation per Cell

For each income × age cell, the GVC is:

- For each rate tier r ∈ {3.5%, 4%, 6%, 8%, 10%}:
  - GVC_r = min(income_mid × rate_r × match_rate, max_gvc)
  - Members_r = total_members × proportion_on_rate_r
- Cell GVC = Σ(Members_r × GVC_r) [summed across rate tiers]
- Members on the non-contributor proportion (ppn) receive $0 GVC

4.3 Self-Employed Members

Self-employed KiwiSaver members (277,986 as at 30 June 2025) are included in the base population throughout. The member counts from Sheet 2 represent all KiwiSaver members regardless of income type, so self-employed members are correctly embedded in the simulation grid in proportion to their age distribution.

4.4 Revised Contribution Rate Proportions

Sheet 14 of the monitoring data provides actual contribution status by income type. Of the 3,405,406 KiwiSaver members, 2,329,951 (68.4%) contributed in 2025:

Income type	Contributing	Not contributing	Total	% contributing
PAYE	2,010,510	451,594	2,462,104	81.7%
Self-employed	169,862	108,124	277,986	61.1%
Passive	101,522	476,953	578,475	17.5%
Hybrid	25,140	19,157	44,297	56.8%
No income	22,917	19,627	42,544	53.9%
Total	2,329,951	1,075,455	3,405,406	68.4%

The Sheet 15 snapshot captures 1,799,359 members with a rate on record, across all income types. The difference of 530,592 were not captured with a rate at that snapshot point but did contribute over the year per Sheet 14. See Section 4.5 for why this is not exclusively a direct-credit/non-payroll group.

This gives:

```
unmatched_rate = (2,329,951 − 1,799,359) / 3,405,406 = 15.58%
```

Section 4.5 describes how this 15.58% is distributed across the pp3–pp10 rate tiers and subtracted from ppn.

4.5 Income Type and the Unmatched-Contributor Rate Assumption

Section 4.4 derives a single overall unmatched_rate of 15.58% from the gap between Sheet 14 (actual contribution status, recorded by income type) and Sheet 15 (rate-captured snapshot, also by income type). This section makes explicit what that step does, and does not, preserve from the underlying income-type data.

Sheet 14 (Section 4.4 table) shows contribution propensity differs sharply by income type: PAYE members contribute at 81.7%, self-employed at 61.1%, hybrid at 56.8%, no-income members at 53.9%, and passive members at only 17.5%. These are large, genuine differences in underlying behaviour. None of this granularity survives into the simulation grid. Assumption A7 spreads the 530,592 unmatched contributors uniformly across income bands. Assumption A8 splits them across the pp3–pp10 rate tiers using each income type's own Sheet 15 rate mix, rather than assigning all of them to the single 3.5% tier; a partial improvement that still can't vary by income band, since Sheet 14's income-type breakdown isn't crossed with income band. This is distinct from the uptake slider's flat 3.5% assumption for hypothetical new contributors (Limitation L6 and Section 5.2), this section concerns the 2,329,951 members who contributed in 2025 across the whole population (Section 4.4), not the smaller age-and-income-eligible subset (2,056,821, Section 7.2).

4.5.1 Rate-Tier Distribution by Income Type

Sheet 15 of the monitoring data cross-tabulates contribution rate by income type. The table below shows each income type's own contribution rate-tier mix among its Sheet 15 rate-captured members. The direct input to Assumption A8's split.

Income type	3% (modelled at 3.5%)	4%	6%	8%	10%
PAYE	64.29%	17.29%	6.94%	5.68%	5.80%
Self-employed	70.32%	12.02%	6.57%	4.76%	6.33%
Passive	71.28%	11.12%	6.05%	4.56%	6.99%
Hybrid	62.36%	15.59%	6.07%	7.26%	8.72%
No income	79.97%	8.19%	5.29%	1.98%	4.57%

4.5.2 Derivation of the Unmatched-Contributor Rate-Tier Split (Assumption 8)

Assumption A8's tier split (66.70% / 15.18% / 6.71% / 5.31% / 6.10%) is derived in two steps. First, each income type's "unmatched" count — the gap between Sheet 14's contribution total and Sheet 15's rate-captured total, by type:

Income type	Sheet 14 (contributed)	Sheet 15 (rate-captured)	Unmatched
PAYE	2,010,510	1,682,340	328,170
Self-employed	169,862	46,752	123,110
Passive	101,522	36,369	65,153
Hybrid	25,140	17,053	8,087
No income	22,917	16,845	6,072
Total	2,329,951	1,799,359	530,592

Second, applying each type's own rate-tier mix (item above) to that type's own unmatched count, not a blended national mix, and summing across types:

Income type	Unmatched	-> pp3	-> pp4	-> pp6	-> pp8	-> pp10
PAYE	328,170	210,993	56,737	22,773	18,649	19,019
Self-employed	123,110	86,571	14,796	8,084	5,864	7,794
Passive	65,153	46,441	7,243	3,939	2,972	4,557
Hybrid	8,087	5,043	1,260	491	587	705
No income	6,072	4,856	497	321	120	278
Total	530,592	353,904	80,533	35,608	28,192	32,353
Total (as %)	100%	66.70%	15.18%	6.71%	5.31%	6.10%

> *Final split matches Assumption A8. This depends on the assumption that unmatched members within an income type share that type's own rate-tier habits (e.g. an unmatched PAYE member is assumed to behave like a rate-captured PAYE member, not like a self-employed member), a reasonable basis given the available data, but independently verifiable from Sheet 14/15 alone.*

This 530,592 figure is referred to as "unmatched contributors" throughout this document, rather than "non-PAYE contributors," because it is not exclusively non-PAYE: comparing Sheet 14 against Sheet 15 by income type shows 328,170 of the 530,592 (62%) are themselves PAYE members whom the rate snapshot simply missed (most plausibly a job change or employment gap around the reference pay period), with the remainder genuinely self-employed, passive, hybrid, or no-income. Each type is weighted using its own Sheet 15 rate mix regardless, see the split in Assumption A8, so this labelling correction doesn't change any calculation, only the term applied to the group.

4.6 Targeting and Uptake-Weighting Parameters

calc_gvc_2026() supports four independent, composable parameters (income targeting, age targeting, uptake weighting, and a minimum contribution rate) allowing a boosted match rate and cap (or boosted uptake, or an adjusted minimum rate) to apply to a subset of the population rather than economy wide. All four default to off/NULL/3.5% and reproduce the $491.4m 2026 default baseline (Section 7.2).

Income targeting: members in income bands with income below a chosen threshold receive a separate boosted match rate and cap; members at or above the threshold continue to receive the base rate and cap. The threshold is compared against each band's income after any income-growth setting is applied (the grown midpoint). Under an income-growth scenario a member whose income rises above the threshold moves out of the boosted group; with growth switched off this is simply the band midpoint.

Age targeting: members in a selected subset of the currently eligible age bands receive a separate boosted match rate and cap; the remaining eligible bands continue to receive the base rate and cap. The UI constrains the age-targeting checkbox list to only offer bands already ticked as eligible, so a band cannot be boosted without first being made eligible.

Uptake weighting: the single uptake_pct parameter (Section 5.2) can be replaced by two tiers, uptake_pct_below and uptake_pct_above, split at a separate income threshold, again compared against grown income (consistent with the income-targeting threshold above). New contributors created in both tiers join the pp3 tier at whatever rate min_rate is set to (default 3.5%), consistent with the mechanism described in the "Minimum contribution rate" paragraph below; only the percentage of non-contributors who take part varies by tier, not the rate they join at.

Precedence when income and age targeting overlap: income targeting is applied first; age targeting is then applied on top and overwrites the effective match rate and cap for any member in a boosted age band, regardless of what income targeting set. So, for a member who qualifies for both an income-targeted boost and an age-targeted boost, the age-targeted settings win. This is deliberate application order, not an accident; verified computationally: combining a 100c/$700 income boost below $40,000 with a 50c/$260.72 age boost for 18–24 leaves every 18–24 member below $40,000 on the 50c age-targeted rate, not the 100c income-targeted rate.

Minimum contribution rate: calc_gvc_2026()'s fourth parameter, min_rate, is exposed in the interface as a "Member minimum contribution rate" slider (2%–10%, default 3.5%). It shifts the rate applied to the pp3 tier only; both existing minimum-rate contributors and any new uptake contributors created by the uptake slider (Section 5.2) join at whatever rate min_rate is set to. Members already on the 4%, 6%, 8%, or 10% tiers are unaffected, consistent with the mechanism described in Section 4.4. At the default 3.5%, this is consistent with the $29,797 cap-reach income, the $1,042.88 contribution needed for a full GVC, and Assumption A6 referenced elsewhere in this document.

Reachability of the maximum: the tool reports the share of contributing members who receive the full maximum government contribution rather than the proportional amount. A member is counted as reaching the maximum where income multiplied by that member's own contribution rate and by the effective match rate is at or above the effective maximum. The calculation runs across all five rate tiers rather than the minimum rate alone, weighting each tier by its proportion of the band, so members on 4%, 6%, 8%, and 10% are counted at their own rate. The denominator is contributing members only; non-contributors are excluded. Where income or age targeting is active, the measure is restricted to the group receiving the enhanced contribution, being members below the income threshold or in boosted age bands, since that is the group a targeted design is intended to reach. With no targeting active it covers all age and income eligible contributors. Because the calculation reads the effective match rate and effective maximum after any targeting has been applied, it does not need to know which lever produced them, and the precedence rule described above carries through automatically. At the 2026 default settings, 83.1% reach the full $260.72 maximum.

Coverage and recipients: a recipient is a member who is age eligible, income eligible, and contributing. Coverage is the count of those members, and the change in coverage is the scenario count minus the 2026 default count. Coverage moves only through the three levers that change who contributes or who qualifies: age eligibility, income eligibility, and non-contributor uptake. It does not move with the match rate or the maximum, which change how much eligible contributors receive rather than how many receive anything. Average GVC per contributing member is total scenario cost divided by this same recipient count, so it is an average over recipients rather than over the whole eligible population. The By Income Band and By Age Band tables report both that measure and an average across all eligible members, including non-contributors, so the two can be read against each other.

For reference, three worked examples illustrate the scale of these levers, all assuming min_rate at its 3.5% default and remaining parameters at their 2026 defaults: income targeting (boosting to 50c/$260.72 below $40,000) gives $511.1m; age targeting (boosting 18–24 to 50c/$260.72) gives $496.5m; uptake weighting (40% uptake below $40,000, 0% above) gives $526.0m.

5. Interactive Model Features

The model is delivered as an interactive R/Shiny application. A sidebar on the left holds all policy controls, and a toggle at the top switches the whole tool between the 2024 observed baseline and the 2026 simulation. In 2026 mode the controls are grouped into five collapsible panels along with a "Reset all to 2026 defaults" button:

1. GVC Settings (match rate, maximum contribution, minimum contribution rate)
2. Income Settings (eligibility cap and income targeting)
3. Age Settings (eligible bands and age targeting)
4. Non-Contributor Uptake
5. Income Growth

The results are presented across five tabs:

Summary: three panels shown together: the *2024 Observed* result (IRD data, fixed), the *2026 Default Baseline* (default policy, no user inputs), and *Your Scenario*, which is reactive to the sidebar and reports the change against the 2026 default. When no inputs have been changed from default, the scenario panel is greyed out.

The Your Scenario panel opens with a one-line summary of every parameter changed from the default, followed by up to seven metric cards:

1. Scenario GVC cost
2. Change against the 2026 default, in dollars and percent
3. Average GVC per contributing member: scenario cost divided by the number of members receiving a contribution (Section 4.6)
4. Change in coverage: the change in the number of members receiving a contribution, against the 2026 default, in members and percent (Section 4.6)
5. Income needed to reach the maximum, calculated as the effective maximum divided by the product of the minimum contribution rate and the effective match rate. Where income or age targeting is active this uses the boosted match rate and maximum, so the card describes the enhanced group rather than the base settings
6. Share of the enhanced group receiving the full maximum (Section 4.6). With no targeting active, this is the share of all eligible contributors
7. New contributors, shown wherever an uptake setting is applied, whether uniform or split at an income threshold

By Income Band: for whichever policy view is selected, GVC cost and member counts grouped into the reporting income bands. Under the 2026 view the table reports total eligible members, contributing members, non-contributing members, the change in contributors against the 2026 default, GVC in millions, average GVC per total eligible member, average GVC per contributor, and each band's share of total GVC, with a total row. The cost chart overlays the 2026 default as a dashed reference line so that the scenario and the default can be read together.

By Age Band: the same measures, columns, total row, and default reference line, grouped by age band.

Distributional Impacts: the user's scenario compared against the 2026 default: green where a group receives more GVC than under the default, red where less, shown by income band and by age band. Both tables report scenario GVC, default GVC, the change in dollars and percent, scenario contributors, default contributors, and the change in coverage, with a total row.

Model Validation: the reproduction checks (2024 baseline, 2026 grid totals, and the default-baseline backward compatibility check) and the full list of assumptions A1 – A10.

The below table lists the sidebar controls, their 2026 default values, and their effect on the calculation.

5.1 Policy Parameter Sliders

Control	Default	Effect
Match rate	0.25	Changes GVC per dollar contributed. Higher rate = higher cost. The slider ranges from $0.25 to $2.00 (step $0.25); the base match rate cannot be set below the 2026 default via this control.
Max GVC cap	$260.72	Caps the GVC per member per year. Entered as a numeric input

Minimum contribution rate	3.5% (slider: 2%–10%)	Shifts the rate applied to pp3-tier members and any new contributors created by the uptake setting only. Members already on higher tiers are unaffected.
Income eligibility cap	$180,000 (entered via a $10k stepped slider)	Also includes a “no cap – include above $180k” toggle. Members above the cap receive $0 GVC; under an income-growth scenario, members whose grown income rises above the cap lose eligibility. The slider stops at $180k because the data has no resolution above it; “No cap” brings in the single “Over $180k” group (approx. 120k members, valued at the $200k assumption in A9). Reporting bands run in uniform $10,000 steps to $180,000 so that the eligibility cap, which also moves in $10,000 increments, always lands on a reporting band edge rather than inside a band. The underlying data is in $1,000 bands throughout this range, so reporting at $10,000 resolution neither loses nor invents detail.
Eligible age bands	16–17 through 55–64 ticked	Members in unticked age bands receive $0 GVC. Bands need not be contiguous, enables age-cohort targeting.

5.2 Non-Contributor Uptake Slider

The uptake slider (0–100%) models a behavioural response: the percentage of current non-contributors assumed to start contributing at the minimum contribution rate (default 3.5%; adjustable 2%–10%, see Section 4.6; see also Section 4.5 for why 3.5% specifically at default, and how this relates to the unmatched-contributor rate assumption applied to the existing contributor base).

The panel offers two modes. By default, a single slider (0-100%, in 5% steps) applies one uptake rate to every non-contributor. Ticking “Different uptake by income” replaces it with three controls: an income threshold, entered on a $10,000 step slider running from $10,000 to $180,000 and matching the income-targeting threshold, and separate uptake percentages below and above that threshold. The threshold is compared against each band’s grown income midpoint, consistent with the income-targeting threshold described in Section 4.6. New contributors created in either tier join the pp3 tier at the minimum contribution rate; only the share of non-contributors who take part varies by tier, not the rate they join at.

The table below shows the range produced by the single-slider mode at the minimum rate’s 3.5% default; each 10 percentage points of uniform uptake adds roughly $13.3m.

Non-contributor uptake	Total 2026 GVC (match/cap at default)
0%	$491.4m
5%	$498.0m
10%	$504.7m
15%	$511.4m
20%	$518.1m
25%	$524.7m
30%	$531.4m
40%	$544.8m
50%	$558.1m
60%	$571.5m
70%	$584.8m

80%	$598.1m
90%	$611.5m
100%	$624.8m

> *The $491.4m baseline reflects observed 2025 contribution behaviour. The uptake slider represents additional behavioural change. It is a sensitivity tool, not a forecast. The figures above assume uniform uptake across all income bands.*

5.3 Income Growth Toggle

When enabled, a 3.5% per annum compound growth rate is applied to the income midpoint of each cell. More members cross the ~$29,797 income at which the GVC cap binds; and, because income eligibility is assessed on the grown income, members whose income rises above the eligibility cap lose eligibility and drop to $0. This second effect reduces cost at the top of the eligible range and can outweigh the first. E.g. two years of 3.5% income growth lifts roughly 32,000 members over the $180k cap (about -$7.4m cost reduction), so that scenario's net effect is a small decrease in total GVC rather than an increase.

> *Income growth is applied to income midpoints within existing bands rather than redistributing member counts across bands. See Limitation L2 in Section 6.*

5.4 Distributional Impacts

Compares the user's scenario against the 2026 default policy settings. Green bars indicate a group receives more GVC than under the default; red bars indicate less.

6. Key Assumptions and Limitations

6.1 Assumptions

Ref	Assumption	Rationale
A1	2024 baseline uses directly observed IRD average GVC per income and age band.	Ensures exact reproduction of $1,022,341,210 total.
A2	KiwiSaver age band member totals are taken directly from Sheet 2 of the 2025 monitoring data.	Anchors member totals to the most reliable available source, avoiding rounding gaps from population-based derivation.
A3	IRD taxable income distribution is used only as a proportional shape within each age band; it does not determine totals.	Separates the role of the two data sets cleanly: monitoring data for totals, taxable income data for within-band shape.
A4	The 15–19 age band is split equally across five ages (1/5 per year) to map into KiwiSaver age bands.	No finer published split is available.
A5	PAYE contribution rate mix from Sheet 6 (2025) is applied by $10k income band.	Most recent available data.
A6	Members currently on the 3% rate move to the minimum contribution rate under the 2026 policy (default 3.5%, adjustable 2%–10%, Section	The 2026 policy raises the minimum from 3% to 3.5%.

	4.6); this assumption holds as stated at the slider's default setting.	
A7	Unmatched contributors (530,592, a mix of PAYE members missed by the Sheet 15 snapshot and genuinely self-employed, passive, hybrid, and no-income contributors; see Section 4.5) are distributed uniformly across income bands using the 15.58% overall rate. See Section 4.5.	Sheet 14 provides a breakdown by income type (Section 4.5) but not crossed with income band, so the 15.58% rate cannot be allocated by band with any better basis.
A8	Unmatched contributors are split across the pp3–pp10 rate tiers using each income type's own Sheet 15 rate mix (66.70% pp3, 15.18% pp4, 6.71% pp6, 5.31% pp8, 6.10% pp10), rather than assigned entirely to pp3.	This is a national average by income type, it still can't vary by income band, since that cross-tab isn't available in Sheet 14/15.
A9	The 'Over $180,000' band uses $200,000 as its midpoint.	This band is ineligible for GVC so the midpoint does not affect cost. The model has no income resolution above $180k; everyone earning more sits in this single band at the assumed $200k midpoint, so it cannot distinguish an $185k earner from a $350k earner. The income-cap slider therefore stops at $180k; the only way to include this group is the “no cap” toggle, which brings the whole group in at once. The $200k figure affects only whether this group is counted as eligible.
A10	Non-contributor uptake is uniform across income bands and age groups, or set separately above and below an income threshold	No empirical basis for differential uptake. Treat slider as sensitivity tool.

6.2 Limitations

Ref	Limitation	Impact
L1	Income data in the 2025 monitoring file reflects the 2024 tax year.	Small income growth effect; partially mitigated by the income growth toggle.
L2	Income growth scales band midpoints; members whose grown income exceeds the eligibility cap correctly lose eligibility (assessed on grown income). Members are not otherwise redistributed across the display bands.	Immaterial below the cap (GVC there is either cap-bound or already captured by the midpoint shift); the material effect, the drop-off at the eligibility cap, is now captured.
L3	The unmatched-contributor rate adjustment (15.58%) is applied uniformly across income bands, and split across rate tiers using each income type's national Sheet 15 rate mix rather than by income band. See Section 4.5.	Self-employed, passive, and hybrid contributors within the same income band are still pooled, since the type-mix can't vary by band. Residual risk is smaller than the flat-3.5% approach, but not eliminated. May still overstate GVC for bands with concentrated passive

		contributors and understate for bands with concentrated self-employed contributors.
L4	The model does not capture employer contributions or voluntary amounts above the rate-implied contribution.	Slightly understates GVC for members contributing above their minimum rate.
L5	Behavioural responses beyond the uptake slider are not modelled.	Static fiscal cost estimate only.
L6	The uptake slider models every newly-contributing member, regardless of true income type, joining at a flat 3.5% rate (Section 5.2). Unlike A8 (Section 6.1), which now varies by income type, the uptake slider still applies a single flat rate to every newly contributing member regardless of type; this is a distinct, unresolved simplification, not the same one as A8.	Uptake scenarios cannot currently distinguish whether newly-contributing members are self-employed, passive, hybrid, or PAYE, understating cost if new contributors skew toward higher rate tiers, overstating it if they skew toward irregular/lump-sum contributors.

7. Model Validation

7.1 2024 Baseline

Check	Model	IRD Observed	Difference
Total GVC (income table)	$1,022,341,210	$1,022,341,210	$0
Total GVC (age table)	$1,022,341,210	$1,022,341,210	$0
Total members	3,348,048	3,348,048	0

7.2 2026 Default Simulation

7.2.1 2026 Default Baseline

Metric	Value
2026 default GVC	$491.4m
2026 default GVC vs 2024	−51.9%
Total age- and income-eligible members	2,933,468
Contributing members (eligible)	2,056,821
Average GVC per contributing member	$238.89
Contributors reaching the full maximum	1,709,870 (83.1%)

7.2.2 Eligible Population by Broad Income Band

The 2,056,821 contributing figure above is a subset of the total age-and-income-eligible population of 2,933,468. The remaining 876,647 are eligible by age and income but do not contribute to

KiwiSaver, so receive $0 GVC. The table below shows how this full eligible population (contributors and non-contributors) are distributed by income band.

Broad income band	Contributors	Non-contributors	Total eligible	% of eligible
$1 - $10k	113,577	301,508	415,085	14.1%
$10k - $20k	124,872	111,497	236,369	8.1%
$20k - $30k	158,177	203,100	361,277	12.3%
$30k - $40k	134,132	80,314	214,446	7.3%
$40k - $50k	165,808	53,505	219,313	7.5%
$50k - $60k	218,601	35,378	253,979	8.7%
$60k - $70k	239,218	23,322	262,540	8.9%
$70k - $80k	205,226	19,316	224,542	7.7%
$80k - $90k	161,543	10,864	172,407	5.9%
$90k - $100k	127,783	7,301	135,084	4.6%
$100k - $110k	103,765	5,692	109,457	3.7%
$110k - $120k	78,630	4,078	82,708	2.8%
$120k - $130k	61,560	3,666	65,226	2.2%
$130k - $140k	48,012	2,992	51,004	1.7%
$140k - $150k	38,520	2,772	41,292	1.4%
$150k - $160k	29,451	4,315	33,766	1.2%
$160k - $170k	24,050	3,524	27,574	0.9%
$170k - $180k	23,898	3,502	27,400	0.9%
Total	**2,056,821**	**876,647**	**2,933,468**	**100.0%**

> *Non-contributing members are concentrated at low incomes. 301,508 of the 415,085 people in the $1 - $10k band (73%) don't contribute, compared to under 15% in every band above $50k. This is not directly comparable to Section 4.2.2's whole population table. Both now use the same reporting bands, but this one excludes under-16s, 65+, and income above $180k, so the same broad bands show smaller totals here.*

The model's $491.4m default sits below Treasury's 2026 BEFU forecast of $560m, reflecting that this model represents observed 2025 contribution behaviour rather than a forward-looking forecast (Section 7.4 discusses other structural reasons the two aren't directly comparable). Section 5.2's uptake sensitivity table gives a sense of scale: each 10 percentage points of uniform non-contributor uptake adds roughly $13.3m, ranging from $491.4m at 0% uptake to $624.8m at 100%. This illustrates how much behavioural change could move the total, it isn't a claimed reconciliation with any external forecast.

7.3 Population Reconciliation

Source	Members	Notes
2025 KiwiSaver monitoring (actual)	3,405,406	Sheet 2 total
Model (scaling approach)	3,405,233	Exact match to Sheet 2 per age band; 173-member residual is the

		excluded 'No Information' age category, not a rounding effect.

> *The model total of 3,405,233 vs the actual 3,405,406 (difference of 173) arises because Sheet 2's "No Information" age category (173 members with unrecorded age) has no equivalent age band in the taxable income distribution shaping data and is therefore excluded from the age-scaling step entirely; a categorical exclusion, not a rounding effect. This is a negligible 0.005% difference with no material effect on GVC estimates.*

7.4 Context: Treasury's Published Policy Costings

Treasury's 2025 Budget Economic and Fiscal Update (BEFU) published fiscal costings for these individual 2026 KiwiSaver policy decisions. They're included here for context and awareness, not as a benchmark this model is expected to reproduce: Treasury's figures are 5-year forecast-period totals, with its own methodology and assumptions, whereas this model produces a single, static, no-growth 2026 year. The two are structurally different kinds of estimate, and comparing them directly, even after annualising Treasury's total, risks implying a precision neither figure supports.

Several factors would reasonably cause the model's estimate of an equivalent scenario to differ from Treasury's, beyond the assumptions and limitations already documented (Sections 6.1–6.2): population growth, including net immigration, grows the underlying KiwiSaver membership base over Treasury's 5-year window in a way this model doesn't attempt to; changes in the workforce, employment and participation rates, wage growth, would shift both who contributes and how much over that period; and, for policy decisions with a paid-parental-leave interaction, Treasury's costings may include a component this model doesn't cover (GVC only). The model's own estimate of each of these scenarios sits in the scenario analysis series rather than repeated here, where it can be read alongside the rest of that scenario work rather than implying a direct benchmark against Treasury.

Lever	Treasury's published estimate (2025 BEFU, 2025/26-2029/30 total)
Halving match rate + cap together	$2,300m
Removing GVC above $180,000	$160m
Extending to 16–17 year-olds	$25m

Treasury "KiwiSaver subsidies" forecasts: the 2025 BEFU gives $545m (2026), $565m (2027), $588m (2028) and $613m (2029); the 2026 BEFU revised 2026 to $560m and extended the series to $583m (2027), $608m (2028), $639m (2029) and $659m (2030).

8. Version History

Version	Key change	Baseline GVC
v3	Single age × income grid; independence assumption	$1,022m (2024 only)
v4	Two separate lookup tables; no simulation	$1,022m (2024 only)
v5	Full 2026 simulation; fine income bands; SE toggle added	$388m
v6	Fixed SE double-count; corrected ppn using Sheet 14	$483m
v7	Member totals from KS monitoring data directly; IRD taxable income distribution data for shape only	$490m

v8	Age eligibility moved from a 16–65 slider to per-band checkboxes (interface only; data pipeline unchanged); added 4.5 non-PAYE rate disclosure, corrected Treasury reconciliation (5.2, 7.2), and added 7.4 lever-by-lever validation against Treasury's published costings	$490m (unchanged)
v9	Added income targeting, age targeting, and uptake weighting as independent, composable parameters to calc_gvc_2026() (4.6); corrected the stale “SE toggle” docstring flagged in v8	$490m (unchanged, verified)
v10	Summary tab redesign (separate 2024 / 2026 default / your-scenario sections) and collapsible accordion sidebar; UI changes only, no calculation changes	$490m (unchanged)
v11	Minimum contribution rate (min_rate) became a user-adjustable parameter (2%–10%, default 3.5%), replacing the previous fixed 3.5% assumption for the pp3 tier and uptake-slider contributors; base match-rate slider range changed from $0.05–$1.00 to $0.25–$2.00 (step $0.25); added a "Reset all to 2026 defaults" control. Several documentation discrepancies surfaced during review; all resolved, including correcting Section 7.2's "Contributing members (eligible)" figure and aligning the app's Model Validation tab assumption numbering (A1–A10) with Section 6.1	$490m (unchanged at default settings)
v12	Completed and extended the v12 fixes across build_dataset_v12.R and shiny_app_v12.R. Build script: corrected get_rate_props() so the $0.01–$100 band maps to the lowest rate tier, and the three IRD taxable income distribution data label overlaps ($20k/$50k/$80k) to the correct $10k tier; replaced the flat-3.5% assumption for unmatched contributors with a type-specific split across pp3–pp10 (66.70/15.18/6.71/5.31/6.10) using each income type’s own Sheet 15 rate mix; renamed the variable set to RATE_CAPTURED_MEMBERS / UNMATCHED_CONTRIBS / UNMATCHED_RATE (the group is a snapshot-vs-outcome gap across all income types, 62% is PAYE, not exclusively non-PAYE). Shiny app: replaced three hardcoded Summary-tab cards with reactive values; scoped members_eff by age AND income eligibility (was age only), so the By Income Band / By Age Band tabs reconcile with §7.2.2 at 2,933,468 and the contributing-eligible figure is 2,056,821; corrected the Model Validation tab’s backward-compatibility check (now expects $491,359,100) and its A7/A8 assumption text (“unmatched contributors”, not “non-PAYE”). New behaviour: income eligibility is now assessed on grown income, so under an income-growth scenario members whose income rises above the cap lose eligibility (the $180k drop-off); the income-targeting and uptake thresholds are likewise compared against grown income. UI: the income-eligibility cap became a $10k-step slider ($10k–$180k) plus a “No cap — include income above $180k” toggle, and the income-targeting and uptake thresholds became matching sliders. Default baseline unchanged.	$491.4m
v13	Reporting income bands moved to uniform $10,000 steps to $180,000 (18 bands plus a single over-$180k group), replacing the previous 14-band grouping, so that the income eligibility cap always lands on a reporting band edge; Tables in Sections 4.2.2 and 7.2.2 rebuilt accordingly, with all retained rows verified unchanged. Summary tab gained average GVC per contributing member, change in coverage, and share of the enhanced group reaching the full maximum; the income-to-reach-maximum card now respects the targeting settings rather than always reporting the base rate and cap. By Income Band, By Age Band, and both Distributional Impacts tables gained a change-in-contributors	$491.4m

	column and a total row, and the cost charts gained a 2026 default reference line. Spinner controls were removed from the three GVC amount inputs, and their $50 floor and $10 step replaced with $0 and $0.01, so a typed value can no longer be snapped onto a coarser grid. Two stale interface strings were corrected: the By Income Band footnote, which still cited 14 reporting bands, and the GVC Settings assumption box, which described the minimum rate tier circularly. Source data files were renamed for external release, and the taxable income distribution correctly attributed to IRD rather than Stats NZ across the build script, app, and documentation. The build script was renamed to Build_dataset_for_GVC_model.R and the app to TAAORC_GVC_Model.R, with version numbering removed from both filenames and headers, so this section is now the sole record of model version. Reachability and coverage defined in Section 4.6. No calculation changes; the default baseline is unchanged and was independently reverified at $491,359,100.	

9. File Inventory

File	Type	Description
app.R	R script	Launcher. Sources TAAORC_GVC_Model.R and returns the app object, which is what a hosting platforms needs app.R to evaluate to. Contains no model logic
manifest.json	JSON	Dependency file for hosted development. Records the R version and package versions used to rebuild the environment. Regenerate with rsconnect::writeManifest() if the packages change
README.md	Markdown	Repository overview: model purpose, levers, outputs, data sources, limitations, repository layout, and deployment instructions.
Build_dataset_for_GVC_model.R	R script	Builds the three CSV datasets from raw source files
TAAORC_GVC_Model.R	R/Shiny script	Interactive modelling application
dataset_by_income.csv	CSV (8 rows)	2024 baseline by income band
dataset_by_age.csv	CSV (7 rows)	2024 baseline by age band

dataset_2026.csv	CSV (1456 rows)	2026 simulation grid (income × age × rate proportions)
2024_IRD_Administrative_KiwiSaver_Data.xlsx	Source data	IRD observed GVC data, 2024 tax year
IRD_KiwiSaver_Monitoring_Project_Data_at_30_June_2025.xlsx	Source data	KiwiSaver monitoring, 30 June 2025
IRD_Taxable_income_distribution_of_individuals_2025.xlsx	Source data	IRD taxable income distribution by age, 2025

In the published repository, app.R, TAAORC_GVC_Model.R, README.md, manifest.json and the three CSV files sit together at the top level, with the build script and the three source spreadsheets in a build subfolder and the published documents in a docs subfolder. The app reads the three CSV files by relative path, so those files must remain alongside it; subfolders are not read. The build script requires the three source spreadsheets in its own working directory and does not need to be run in order to use the app.